

Robust Tensor Decomposition Under Multi-Mode Outlier Corruptions

Yuning Qiu , Guoxu Zhou , Andong Wang, Zhenhao Huang , Qibin Zhao , and Shengli Xie , *Fellow, IEEE*

Abstract—Robust tensor decomposition methods have emerged as effective tools for handling corrupted tensor data. However, conventional approaches require prior knowledge to determine the specific mode in which outliers are distributed. Moreover, these approaches are inherently restricted to exploring outliers within a single mode, which proves inadequate for high-order tensors where outliers often manifest across multiple modes simultaneously. This paper proposes a novel multi-mode outlier-robust tensor ring decomposition (ORTRD) framework to address this limitation. Our approach introduces a multi-mode tensor structure sparsity measure that effectively characterizes outliers distributed across multiple modes, eliminating the need for prior knowledge on outlier distribution patterns. By integrating this sparsity measure with low-rank tensor ring representation, our framework captures both the underlying low-rank structure and complex outlier patterns. We develop an efficient multi-block ADMM algorithm to solve the resulting optimization problem and establish theoretical convergence guarantees under mild conditions. Furthermore, we provide rigorous analysis of both deterministic and non-asymptotic bounds for the estimation error, theoretically validating the recovery performance of the proposed approach. Extensive experiments on synthetic data, hyperspectral videos, and light field images demonstrate the superior performance of our method in handling various types of outlier corruptions compared to state-of-the-art approaches.

Index Terms—Tensor network, robust tensor decomposition, tensor ring decomposition, tensor recovery.

I. INTRODUCTION

TENSORS are mathematical representations of multidimensional data that generalize vectors and matrices to higher orders. While these high-order structures naturally encode multiway interactions in data, extracting interpretable patterns and relationships from them poses significant challenges. To overcome these challenges, tensor decomposition has emerged as a fundamental tool for analyzing multidimensional data across various scientific domains [1]. Unlike traditional matrix-based methods, tensor decomposition factors high-order tensors into the multilinear product of multiple compact factors. Therefore, tensor decomposition not only reduces parameter complexity but also reveals latent multilinear patterns that might be lost in matrix-based analyses. This capability is particularly valuable in applications such as signal processing [2], [3], [4], [5], computer vision [6], [7], [8], and machine learning [9], [10], [11], [12].

However, these conventional tensor decomposition methods are inherently vulnerable to sparse noise, as they typically assume Gaussian noise in the data. When confronted with sparse but significant corruptions or outliers, these methods often fail to recover the underlying low-rank structure accurately. To address these limitations, robust tensor decomposition (RTD) methods [13], [14], [15], also known as tensor robust principal component analysis (TRPCA), have been developed and extensively studied in recent years. Unlike conventional approaches, RTD explicitly models corruption as a combination of dense noise and sparse outlier components. This modeling strategy allows RTD to capture the intrinsic low-rank tensor structure from outlier corruptions effectively. The effectiveness of RTD has driven remarkable success across diverse applications, including robust feature extraction [16], [17], image processing [18], [19], video background subtraction [20], [21], and anomaly detection [22], [23], [24], achieving consistently excellent performance.

The fundamental principle of RTD lies in simultaneously separating a corrupted tensor into its low-rank structure and sparse components. Recent works primarily focused on strategies for capturing the low-rank structure within high-order tensor data, while characterizing outlier tensors still presents a fundamental challenge. Different from vector or matrix data, the outliers may contain specific patterns aligned with high-order tensors. As illustrated in Fig. 1, outliers in a third-order tensor can manifest in three distinct sparsity patterns: element-wise, fiber-wise, and slice-wise, respectively. Therefore, existing methods

Received 18 April 2025; revised 15 February 2026; accepted 4 April 2026. Date of publication 15 May 2026; date of current version 8 August 2026. This work was supported in part by the Natural Science Foundation of China under Grant 62203124 and Grant 62103110 and in part by the Guangdong Natural Science Foundation under Grant 2023A1515012916 and Grant 2024A1515010114. Recommended for acceptance by Z. Wang. (Corresponding author: Guoxu Zhou.)

Yuning Qiu is with the School of Automation, Guangdong University of Technology, Guangzhou 510006, China, and also with the Center for Advanced Intelligence Project (AIP), RIKEN, Tokyo 351-0198, Japan (e-mail: yuning.qiu.gd@gmail.com).

Guoxu Zhou is with the School of Automation, Guangdong University of Technology, Guangzhou 510006, China, and also with the Key Laboratory of Intelligent Detection and The Internet of Things in Manufacturing, Ministry of Education, Guangzhou 510006, China (e-mail: gx.zhou@gdut.edu.cn).

Andong Wang is with the Center for Advanced Intelligence Project (AIP), RIKEN, Tokyo 351-0198, Japan (e-mail: andong.wang@riken.jp).

Zhenhao Huang is with the School of Automation, Guangdong University of Technology, Guangzhou 510006, China (e-mail: zhhuang.gdut@qq.com).

Qibin Zhao is with the Center for Advanced Intelligence Project (AIP), RIKEN, Tokyo 351-0198, Japan, and also with the School of Automation, Guangdong University of Technology, Guangzhou 510006, China (e-mail: qibin.zhao@riken.jp).

Shengli Xie is with the School of Automation, Guangdong University of Technology, Guangzhou 510006, China, and also with the Guangdong-HongKong-Macao Joint Laboratory for Smart Discrete Manufacturing, Guangzhou 510006, China (e-mail: shlxie@gdut.edu.cn).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TKDE.2026.3687121>, provided by the authors.

Digital Object Identifier 10.1109/TKDE.2026.3687121

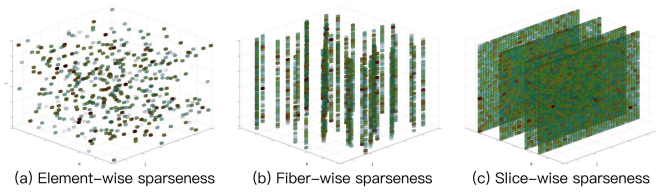


Fig. 1. Illustration of different sparsity patterns in tensor data: (a) element-wise sparsity, where individual entries are sparse; (b) fiber-wise sparsity along mode-1 (column) direction; and (c) slice-wise sparsity along the lateral direction.

usually characterize outlier tensors through two main approaches: element-wise sparsity [14], [25], [26], which treats outliers as independent corruptions similar to matrix or vector data, and structured sparsity, which captures outlier patterns through fiber-wise [27] or slice-wise [28] measures along one specific direction. While these approaches have shown effectiveness in certain scenarios, they are fundamentally constrained by the underlying assumptions about outlier directions. Specifically, element-wise sparsity fails to leverage the inherent structure of outlier patterns, whereas direction-specific structure sparsity measures necessitate prior knowledge about the corrupted direction. However, in real-world data, we often cannot determine which directions are affected by outliers or how many modes simultaneously contain outlier corruptions.

Motivated by these challenges, we propose a novel multi-mode outlier-robust tensor ring decomposition (ORTRD) framework. Our approach introduces a novel multi-mode tensor structure sparsity that can effectively characterize outliers distributed across multiple modes simultaneously (Definition 6), eliminating the need for prior knowledge on corruptions. The proposed framework integrates this sparsity measure with tensor ring (TR) decomposition to capture the underlying low-rank structure, leveraging TR's circular shifting invariance property for enhanced representation capability. The optimization problem is addressed through an efficient multi-block ADMM algorithm, with theoretical convergence guarantees established under mild conditions (Theorem 1). To theoretically validate our approach, we establish both deterministic and non-asymptotic upper bounds for the estimation error (Theorem 2 and Theorem 3), providing rigorous guarantees for the recovery performance. Corollary 1 further extends Theorem 3 by relaxing the Gaussian noise assumption to the more general subexponential noise. In summary, our work contributes in the following ways:

- To estimate low-rank and sparse tensors simultaneously, we develop a novel RTD approach by incorporating a newly established multi-mode tensor structure sparsity alongside a TR rank representation, and propose an efficient optimization algorithm with rigorously proven convergence properties to solve the optimization problem.
- We theoretically reveal the non-asymptotic upper bounds for the estimation error. These results provide rigorous guarantees for the recovery performance of the proposed approach under mild conditions, justifying its effectiveness in handling multi-mode outlier corruptions.
- The proposed approach demonstrates superior versatility across various tensor data recovery tasks. Through

extensive experiments on synthetic low-rank tensor data, hyperspectral videos (HSVs), and light field images, we validate its effectiveness in handling different types of outlier patterns.

Some preliminary results of this work appeared in the 38th AAAI Conference on Artificial Intelligence (AAAI 2024) [29]. This journal version significantly extends the conference paper by refining the problem formulation, providing a more complete and rigorous convergence analysis of the proposed algorithm, and substantially enhancing the experimental evaluation in terms of scale, diversity, and depth. These extensions offer a more comprehensive and theoretically grounded understanding of the problem. The remainder of this article is organized as follows. Section I-A presents a comprehensive review of related works in RTD. Section II introduces necessary notations and preliminaries for tensor operations. Section III details the proposed ORTRD framework, including the mathematical model formulation, optimization algorithm development, convergence analysis, and computational complexity analysis. In Section IV, we provide a theoretical analysis of the recovery performance with both deterministic and non-asymptotic bounds. In addition, we also analyze the statistical performance under a more general subexponential noise. Section V presents extensive experimental results on both synthetic and real-world data to demonstrate the effectiveness of the proposed approach. Finally, Section VI concludes the paper with a summary of our contributions and discusses potential directions for future research.

A. Related Works

1) *Sparsity Characterization in Tensors*: The characterization of sparse corruptions in high-order tensors has been addressed through various measures. The most straightforward approach is to describe tensor outliers using element-wise sparsity via the ℓ_1 norm [14], [15], [25], [30], [31], [32], [33], owing to its natural extension from vectors and matrices to tensors. However, this approach proves inadequate when outliers exhibit structured patterns [34], [35], [36], particularly when corruptions concentrate along specific fibers rather than distributing uniformly across the tensor. To address structured corruption, recent approaches have been developed based on mode-specific sparsity measures. One prominent strategy involves measuring fiber-wise sparsity by applying the $\ell_{2,1}$ norm to one specific direction [26], [27], [28], [32]. This framework has been further extended to handle slice-wise outlier corruptions through the $\ell_{F,1}$ norm [28], [37], effectively capturing matrix-valued outliers across specific modes. Nevertheless, these structured sparsity approaches face a fundamental limitation: they all require prior knowledge about the specific modes where outliers are distributed before applying the appropriate sparsity measure ($\ell_{2,1}$ or $\ell_{F,1}$ norm) along those modes, making them inherently dependent on accurate outlier mode identification.

2) *Tensor Ring (TR) Decomposition*: Unlike matrices where rank has a unique definition, tensors admit multiple rank concepts due to the diversity of tensor decomposition frameworks. The CP rank, which extends from matrix SVD most directly, defines rank as the minimum number of rank-one tensor

components [38], [39]. However, the NP-hardness of CP rank estimation significantly limits its practical utility. The Tucker rank offers a multilinear generalization, representing rank as a vector of mode-wise ranks to accommodate the multi-dimensional nature of tensors [40]. Nevertheless, Tucker decomposition often suffers from the curse of dimensionality. The tensor train (TT) decomposition addresses these limitations by representing a tensor using the sequential contraction of multiple third-order core tensors [41]. Building upon this, the tensor ring (TR) decomposition further enhances low-rank estimation capability by achieving circularly shifting invariant contraction of core tensors, enabling more balanced representations across all modes [42]. This advantage has led to successful applications of TR decomposition across various domains, including tensor completion, denoising, and robust tensor decomposition. The recently introduced fully connected tensor networks (FCTN) [43] and tensor wheel (TW) decomposition [44] can capture long-range inter-mode dependencies beyond TR/TT, making it a promising backbone for future robust formulations once a corruption model and tractable convex surrogates are developed. Recently, the tensor tubal rank [45], using a new tensor algebra by defining the rank in the transformed domain for third-order tensors, has also demonstrated impressive performance in applications involving multiview clustering [46], [47], tensor regression [48], [49], image and video processing [50], [51]. Nevertheless, extending this approach to higher-order tensors remains an open and actively explored research problem.

3) *Robust Tensor Decomposition (RTD)*: RTD extends the fundamental concept of RPCA to the tensor domain. This extension presents substantial theoretical and computational challenges, primarily due to the increased complexity of tensor rank and sparsity formulations compared to their matrix counterparts. In characterizing the low-rank component, RTD has evolved through increasingly advanced tensor-rank formulations, each capturing distinct low-rank structure. This evolution began with Tucker rank as a direction generation of higher order RPCA (HoRPCA) [14], [52], before advancing to more effective approach such as robust Bayesian CP decomposition (BRTF) [53], TT robust principle component analysis (TTRPCA) [54], [55] for higher-order tensors, and robust TR completion (RTRC) [56] for circular dependent data, thus providing flexible and effective frameworks for solving the RTD problem. While low-rank modeling has advanced considerably, the development of effective methods for modeling the sparse component has remained far behind. Current approaches mainly rely on the element-wise sparse pattern similar to vector sparsity and matrix sparsity. Some recent research works have extended it to structure sparsity, i.e., fiber-wise [26], [27], [28], [32] or slice-wise sparsity [28], [37], for tensor data. However, these structural sparsity measures require the user to predefine which mode is corrupted, i.e., they rely on prior information that is often unavailable for tensor data. Moreover, when outliers simultaneously occur across several modes, these structural sparsity measures break down since they cannot identify or demix the outliers from multiple directions. Our work introduces MTSN, a novel convex penalty that explicitly models multi-mode outliers without acquiring prior knowledge of the corrupted modes, integrates it into a robust

TABLE I
SUMMARY OF NOTATIONS USED IN THIS ARTICLE

Notation	Description
x	Scalar
$\mathbf{x}$	Vector
$\mathbf{X}$	Matrix
$\mathcal{X}$	Tensor
$\mathcal{X}(i_1, i_2, \dots, i_K)$	$(i_1, i_2, \dots, i_K)$ th entry of tensor
$[K]$	Set of integers $\{1, 2, \dots, K\}$
$\ \mathbf{X}\ _\infty$	ℓ_∞ norm of tensor $\mathcal{X}$
$\ \mathbf{X}\ _*$	Nuclear norm of matrix
$\ \mathbf{X}\ _{2,1}$	$\ell_{2,1}$ norm of matrix
$\ \mathbf{X}\ _{2,\infty}$	Dual norm of $\ell_{2,1}$ norm
$\ell_{2,1}^k(\mathcal{X})$	Mode- k Fiber-wise $\ell_{2,1}$ norm of tensor $\mathcal{X}$
$\mathbf{X}_{(k)}$	Mode- k unfolding of tensor $\mathcal{X}$
$\mathbf{X}_{\langle k \rangle}$	Mode- k circular unfolding of tensor $\mathcal{X}$
$c_1, c_2, \dots$	Universal constants

tensor decomposition framework, and provides non-asymptotic estimation-error bounds together with convergence guarantees for the multi-block ADMM.

II. NOTATIONS AND PRELIMINARIES

We begin this section by presenting the notations outlined in Table I, followed by fundamental definitions, preliminaries and lemmas necessary for our remaining theoretical development.

Definition 1 (Mode- k canonical unfolding [1]): For a K th-order tensor $\mathcal{X}$, we define its mode- k canonical unfolding as a matrix $\mathbf{X}_{(k)} \in \mathbb{R}^{d_k \times \prod_{j \neq k} d_j}$. Each element of this matrix representation is mapped from the original tensor through:

$$\mathbf{X}_{(k)}(\overline{i_k, i_1 \dots i_{k-1} i_{k+1} \dots i_K}) = \mathcal{X}(i_1, i_2, \dots, i_K), \quad (1)$$

where $\overline{i_1 \dots i_{k-1} i_{k+1} \dots i_K}$ is the multi-index operation introduced in [57].

Definition 2 (TR decomposition [42]): A K th-order tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$ admits a tensor ring (TR) structure if it can be expressed as a circular contraction of K core tensors $\{\mathcal{G}^{(k)}\}_{k=1}^K$, where each core $\mathcal{G}^{(k)} \in \mathbb{R}^{r_k \times d_k \times r_{k+1}}$ is a third-order tensor with $r_{K+1} = r_1$. The relationship between the original tensor and its cores is given by:

$$\mathcal{X}(i_1, i_2, \dots, i_K) = \sum_{u_1, u_2, \dots, u_K} \prod_{k=1}^K \mathcal{G}^{(k)}(u_k, i_k, u_{k+1}), \quad (2)$$

where $[r_1, r_2, \dots, r_K]$ represents the TR ranks characterizing the complexity of the decomposition.

To effectively address lowrank TR structures, circular tensor unfolding provides a robust framework for applying matrixanalysis techniques.

Definition 3 (Tensor Circular Unfolding [58], [59]): Given an arbitrary K th-order tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$, its mode- k circular unfolding matrix is $\mathbf{X}_{\langle k \rangle} \in \mathbb{R}^{d_{1,k} \times d_{2,k}}$, where $d_{1,k} = \prod_{u=l+1}^{k-1} d_u$ and $d_{2,k} = \prod_{u=k}^l d_u$, and $s = \lceil K/2 \rceil$ denotes the number of indexes maintain in $d_{2,k}$, and l is given by

$$l = \begin{cases} k + s - 1, & k + s \leq K, \\ k + s - 1 - K, & \text{otherwise.} \end{cases} \quad (3)$$

Alternatively, its element-wise form is denoted by $\mathcal{X}(i_1, i_2, \dots, i_K) = \mathbf{X}_{(k)}(\overline{i_{l+1}, \dots, i_{k-1}}, \overline{i_k}, \dots, \overline{i_l})$.

Lemma 1: [58], [59] Given a K th-order tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$ with TR rank $[r_1, r_2, \dots, r_K]$, the rank of circular unfolding matrix $\mathbf{X}_{(k)}$ is bounded by $r_k r_{k+s}$.

Leveraging Lemma 1, we can convert the intractable TR-rank minimization into K separate matrix-rank minimization problems. Replacing the nonconvex rank function with its convex surrogate—the matrix nuclear norm yields the tensor-ring nuclear norm (TRNN) [58], [59], which serves as an effective means of exploiting low-rank TR structure across diverse applications.

Definition 4: [58], [59] Given a K th-order tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$, its TRNN is defined by $\|\mathcal{X}\|_{trnn} := \sum_{k=1}^K \alpha_k \|\mathbf{X}_{(k)}\|_*$, where $\alpha_k > 0, k \in [K]$ and $\sum_{k=1}^K \alpha_k = 1$ denotes the weight parameter.

Lemma 2: [60] Given an arbitrary K th-order tensor $\mathcal{X}$, the dual norm of its TRNN is given by $\|\mathcal{X}\|_{trnn}^* = \inf_{\mathcal{X}=\mathcal{X}^1+\mathcal{X}^2+\dots+\mathcal{X}^K} \max_{k \in [K]} \alpha_k^{-1} \|\mathbf{X}_{(k)}^k\|$, where $\|\cdot\|$ denotes the matrix spectral norm, and $\mathbf{X}_{(k)}^k$ is the k th latent tensor component.

Lemma 2 plays a crucial role in the analysis of the statistical performance of the proposed approach.

Definition 5 (Mode- k Fiber-wise $\ell_{2,1}$ Norm): Let $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$ be a K -th order tensor. The mode- k fiber-wise $\ell_{2,1}^k$ norm is defined on

$$\ell_{2,1}^k(\mathcal{X}) := \sum_{\mathbf{i}_{\setminus k}} \|\mathcal{X}(i_1, \dots, i_{k-1}, :, i_{k+1}, \dots, i_K)\|_2, \quad (4)$$

where $\mathbf{i}_{\setminus k} = (i_1, \dots, i_{k-1}, i_{k+1}, \dots, i_K)$, and $\|\cdot\|_2$ denotes the ℓ_2 norm on fibers. Alternatively, it can be equivalently expressed through tensor-matrix canonical unfolding as:

$$\ell_{2,1}^k(\mathcal{X}) := \sum_{i_k} \|\mathbf{X}_{(k)}(:, i_k)\|_2 \quad (5)$$

where $\mathbf{X}_{(k)}$ denotes the mode- k canonical unfolding of tensor $\mathcal{X}$.

Note that we place the $\ell_{2,1}$ penalty on the mode- k canonical unfolding rather than on the circular unfolding. This design indicates a modeling distinction. Low rank is a global structure of the clean tensor. In contrast, outliers are typically mode dependent. Using canonical unfoldings ensures that each column corresponds to a single mode fiber, so the $\ell_{2,1}$ norm targets sparsity along mode k only. By comparison, the circular unfolding merges multiple modes into each column, and an $\ell_{2,1}$ penalty there would entangle the combinations of outliers from multiple modes. When $K = 2$, it reduces to the matrix $\ell_{2,1}$ norm [34], [61].

III. MULTI-MODE OUTLIER ROBUST TENSOR RING DECOMPOSITION

A. The Proposed Model

Let $\mathcal{Y} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$ denote the observed tensor corrupted by outliers. We posit that the underlying clean tensor admits a

lowrank representation, while a sparse tensor captures anomalous corruptions distributed across multiple modes. Accordingly, the observation model can be written as:

$$\mathcal{Y} = \mathcal{T} + \mathcal{S} + \mathcal{E}, \quad (6)$$

where $\mathcal{T}$ represents the underlying low-rank component, $\mathcal{S}$ denotes the sparse outlier component, and $\mathcal{E}$ captures the dense noise. Before stating our main results, we first establish a mild condition on the low-rank tensor $\mathcal{T}$. We require the magnitude of entries in $\mathcal{T}$ to be bounded, which naturally holds for many real-world data tensors such as images and videos:

Assumption 1: The low-rank tensor $\mathcal{T}$ is assumed to have a bounded ℓ_∞ norm:

$$\|\mathcal{T}\|_\infty = \varphi < \infty,$$

thereby precluding any spiky behavior.

The outlier patterns in a tensor $\mathcal{S}$ pose distinct modeling challenges. First, many existing methods require the corrupted mode to be specified in advance. This may be reasonable for matrices with a single feature dimension, but tensors typically have multiple feature modes (e.g., spatial and angular in light-field images), so the corrupted mode is usually unknown. Second, when outliers occur simultaneously across several modes, the $\ell_{2,1}$ norm and its direct tensor extensions are inadequate because they cannot identify or demix multi-mode corruptions. To address these issues, we introduce the multi-mode tensor structure norm (MTSN), defined as follows.

Definition 6 (Multi-mode Tensor Structure Norm, MTSN): For a K th-order tensor $\mathcal{S}$, we define the following tensor structural norm:

$$\|\mathcal{S}\|_{mtsn} := \inf_{\mathcal{S}=\mathcal{S}^1+\mathcal{S}^2+\dots+\mathcal{S}^K} \sum_{k=1}^K \beta_k \ell_{2,1}^k(\mathcal{S}^k), \quad (7)$$

Here, the tensor $\mathcal{S}$ is decomposed into K latent components $\{\mathcal{S}^k\}_{k=1}^K$, with $\ell_{2,1}^k(\mathcal{S}^k)$ measuring the structure sparsity of the mode- k unfolding matrix $\mathbf{S}_{(k)}^k$. The positive weights $\{\beta_k\}_{k=1}^K$ balance the contribution of different modes. For simplicity, we usually let $\beta_1 = \beta_2 = \dots = \beta_K$.

From Definition 6, we can see that MTSN represents the outlier tensor as a sum of mode-specific components, where each component captures outliers along one particular mode. The infimum guarantees to search for a latent decomposition that minimizes the sum of K $\ell_{2,1}^k(\cdot)$ as small as possible.

To analyze the statistical performance of recovered performance, we establish theoretical guarantees under a standard incoherence condition. This mild assumption, which has been successfully employed in various tensor decomposition frameworks [15], [62], [63], is formalized as follows:

Assumption 2: For all $l, k \in [K]$ with $l \neq k$, the l th unfolding of each latent outlier component $\mathcal{S}^k$ obeys the bound

$$\|\mathcal{S}_{(l)}^k\|_{2,\infty} \leq \varrho.$$

For a matrix $\mathbf{A}$, $\|\mathbf{A}\|_{2,\infty} = \max_j \|\mathbf{A}(:, j)\|_2$ is the maximum column energy. Thus, Assumption 2 indicates that a mode- k outlier component does not concentrate on mode- ℓ fibers, i.e., its energy is spread across many columns in $(\mathcal{S}^k)_{(l)}$. This is

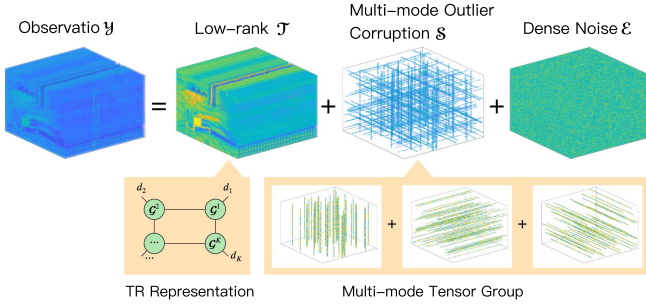


Fig. 2. The problem addressed in this study: recover the “clean” low-rank tensor from multi-mode outlier and dense noise corrupted tensor.

typically satisfied for group-structured corruptions whose active mode- k fibers are dispersed across the other indices. In practice, the bound is mild and scales with data magnitude, since

$$\|(\mathbf{S}_k)_{(\ell)}\|_{2,\infty} \leq \sqrt{d_\ell} \|(\mathbf{S}_k)_{(\ell)}\|_\infty = \sqrt{d_\ell} \|\mathbf{S}_k\|_\infty,$$

so for normalized images/videos one may take $\varrho = O(\sqrt{d_\ell})$.

Assumption 2 is also essential for the identifiability of MTSN, where we present it in the following remark.

Remark 1: Since $\|\mathbf{S}\|_{mtsn}$ is defined via an infimal convolution over mode-wise fiber-sparse models, the decomposition on $\mathbf{S} = \sum_{k=1}^K \mathbf{S}^k$ is not unique in general when $\mathbf{S}$ lies in the overlap (e.g., a single-entry corruption) [64], [65]. In MTSN, assume $\mathbf{S}_{(k)}^k = (\mathbf{S}^k)_{(k)}$ has at most s_k nonzero columns and Assumption 2 holds. Then for any $\ell \neq k$,

$$\begin{aligned} \ell_{2,1}^\ell(\mathbf{S}^k) &= \|(\mathbf{S}^k)_{(\ell)}\|_{2,1} \geq \frac{\|\mathbf{S}^k\|_F^2}{\|(\mathbf{S}^k)_{(\ell)}\|_{2,\infty}} \geq \frac{\|\mathbf{S}^k\|_F^2}{\varrho}, \\ \ell_{2,1}^k(\mathbf{S}^k) &= \|\mathbf{S}_{(k)}^k\|_{2,1} \leq \sqrt{s_k} \|\mathbf{S}^k\|_F, \end{aligned} \quad (8)$$

where $\|A\|_F^2 \leq \|A\|_{2,\infty} \|A\|_{2,1}$ for any matrix A . Therefore, whenever $\|\mathbf{S}^k\|_F / \varrho > (\beta_k / \beta_\ell) \sqrt{s_k}$, then $\beta_k \ell_{2,1}^k(\mathbf{S}^k) < \beta_\ell \ell_{2,1}^\ell(\mathbf{S}^k)$ holds, implying an identifiable mode-wise assignment for non-degenerate fiber corruptions.

Based on the established assumptions and our proposed multi-mode tensor structure norm, we now formulate a comprehensive approach to recover both the low-rank tensor and multi-mode outliers. We regularize the low-rank component using the TRNN and constrain the outlier component through the MTSN, leading to the proposed ORTRD model:

$$\begin{aligned} \min_{\mathcal{T}, \mathcal{S}} \quad & \frac{1}{2} \|\mathbf{Y} - \mathcal{T} - \mathcal{S}\|_F^2 + \lambda \|\mathcal{T}\|_{trnn} + \tau \|\mathcal{S}\|_{mtsn}, \\ \text{s.t.} \quad & \|\mathcal{T}\|_\infty \leq \varphi, \quad \|\mathbf{S}_{(l)}^k\|_{2,\infty} \leq \varrho, l \neq k, \end{aligned} \quad (9)$$

where $\lambda > 0$ and $\tau > 0$ are penalty parameters. This optimization problem can be efficiently solved using the Alternating Direction Method of Multipliers (ADMM) [66], with detailed algorithmic procedures presented in the subsequent section. The graphical illustration for the proposed ORTRD model is given in Fig. 2.

B. Optimization Algorithm

To facilitate efficient optimization, we introduce several methodological refinements. First, we relax the equality constraint by introducing a residual variable $\mathcal{E}$ and decompose the outlier tensor as $\mathcal{S} = \sum_{k=1}^K \mathbf{S}^k$. For computational efficiency, we proceed without the explicit $\ell_{2,\infty}$ constraints on $\mathbf{S}_{(l)}^k, l \neq k$. This leads to the following refined minimization problem:

$$\min_{\mathcal{T}, \{\mathbf{S}^k\}_{k=1}^K, \mathcal{E}} \quad \frac{1}{2} \|\mathcal{E}\|_F^2 + \tau \sum_{k=1}^K \beta_k \|\mathbf{S}_{(k)}^k\|_{2,1}$$

$$+ \lambda \sum_{k=1}^K \|\mathbf{M}_{(k)}^k\|_* + I_T(\mathcal{T}),$$

$$\text{s.t.} \quad \mathcal{Y} = \mathcal{T} + \sum_{k=1}^K \mathbf{S}^k + \mathcal{E}, \mathbf{M}^k = \mathcal{T}, k \in [K], \quad (10)$$

where $I_T(\mathcal{T})$ a indicator function:

$$I_T(\mathcal{T}) := \begin{cases} 0, & \text{if } \mathcal{T} \in I_T, \\ +\infty & \text{otherwise,} \end{cases} \quad (11)$$

where the constraint set $T := \{\mathcal{T} \mid \|\mathcal{T}\|_\infty \leq \varphi\}$.

The augmented Lagrangian function is given by

$$\begin{aligned} \ell_\rho(\mathcal{T}, \{\mathbf{S}^k\}_{k=1}^K, \{\mathbf{M}^k\}_{k=1}^K, \{\mathcal{Q}^k\}_{k=1}^K, \mathcal{P}) \\ = \frac{1}{2} \|\mathcal{E}\|_F^2 + \lambda \sum_{k=1}^K \alpha_k \|\mathbf{M}_{(k)}^k\|_* + \tau \sum_{k=1}^K \beta_k \|\mathbf{S}_{(k)}^k\|_{2,1} \\ + I_T(\mathcal{T}) + \sum_{k=1}^K \langle \mathcal{T} - \mathbf{M}^k, \mathcal{Q}^k \rangle + \frac{\rho}{2} \|\mathcal{T} - \mathbf{M}^k\|_F^2 \\ + \langle \mathcal{Y} - \mathcal{T} - \sum_{k=1}^K \mathbf{S}^k - \mathcal{E}, \mathcal{P} \rangle \\ + \frac{\rho}{2} \|\mathcal{Y} - \mathcal{T} - \sum_{k=1}^K \mathbf{S}^k - \mathcal{E}\|_F^2. \end{aligned} \quad (12)$$

The optimization problem in (12) involves multiple variable blocks, which extend beyond the classical two-block ADMM framework. To ensure robust convergence properties, we enhance our algorithm with a Jacobian ADMM variant by incorporating proximal terms in all subproblems. This modification ensures strict convexity of each subproblem, guaranteeing unique solutions throughout the optimization process. Specifically, we minimize the augmented Lagrangian function for each subproblem:

$$\mathcal{X} = \arg \min_{\mathcal{X}} \ell_\rho(\mathcal{X}) + \frac{\theta}{2} \|\mathcal{X} - \mathcal{X}^t\|_F^2. \quad (13)$$

Then, we can solve each subproblem separately by deriving the closed-form solutions for different variables.

$\mathbf{S}^k$ sub-problem: According to the augmented Lagrangian, the $\mathbf{S}^k$ sub-problem can be formulated as

$$\min_{\mathbf{S}^k} \quad \frac{\theta_k + \rho}{2} \|\mathbf{S}^k - \frac{\rho(\mathcal{Y} - \mathcal{T}^t - \mathbf{S}^{\neq k,t} - \mathcal{E}^t) + \mathcal{P}^t + \theta_k \mathbf{S}^{k,t}}{\theta_k + \rho}\|_F^2$$

$$+ \tau\beta_k \|\mathbf{S}_{(k)}^k\|_{2,1}, k \in [K], \quad (14)$$

where $\mathbf{S}^{\neq k,t} = \sum_{i \neq k} \mathbf{S}^{i,t}$. The closed-form solution can be directly given by the proximal operator:

$$\mathbf{S}_{(k)}^k(:, j) = \text{Prox}_{\tau\beta_k/(\theta_k+\rho)}^{2,1}(\mathbf{Z}(:, j)), \forall j, \quad (15)$$

where $\mathbf{Z} = (\rho(\mathbf{Y} - \mathbf{T} - \mathbf{S}^{\neq k,t} - \mathbf{E}) + \mathbf{P})/(\rho + \theta_k)$ and the proximal operator $\text{Prox}_{\tau\beta_k/(\theta_k+\rho)}^{2,1}(\cdot)$ is given by

$$\begin{aligned} & \text{Prox}_{\tau\beta_k/(\theta_k+\rho)}^{2,1}(\mathbf{Z}_{(k)}(:, j)) \\ &= \left(1 - \frac{\tau\beta_k/(\theta_k+\rho)}{\max(\|\mathbf{Z}_{(k)}(:, j)\|_2, \tau\beta_k/(\theta_k+\rho))}\right) \mathbf{Z}_{(k)}(:, j), \forall j. \end{aligned} \quad (16)$$

$\mathcal{M}^k$ sub-problem: The $\mathcal{M}^k$ sub-problem can be given by

$$\begin{aligned} \min_{\mathcal{M}^k} & \frac{\rho + \theta_{K+1}}{2} \|\mathcal{M}^k - \frac{\rho\mathcal{T}^t + \mathbf{Q}^{k,t} + \theta_{K+1}\mathcal{M}^{k,t}}{\rho + \theta_{K+1}}\|_{\text{F}}^2 \\ & + \lambda\alpha_k \|\mathcal{M}_{(k)}^k\|_* \end{aligned} \quad (17)$$

whose closed-form solution is given by

$$\mathcal{M}^{k,t+1} = \text{Fold}_{(k)}(\text{Prox}_{\lambda\alpha_k/(\rho+\theta_{K+1})}^*(\mathbf{R}_{(k)})), \quad (18)$$

where $\mathbf{R} = (\rho\mathcal{T}^t + \mathbf{Q}^{k,t} + \theta_{K+1}\mathcal{M}^{k,t})/(\rho + \theta_{K+1})$, and $\text{Fold}_{(k)}(\cdot)$ is the folding operation of the unfolding operation given in Definition 3, $\text{Prox}_{\lambda\alpha_k/(\rho+\theta_{K+1})}^*(\cdot)$ denotes the singular value thresholding (SVT) [67] method:

$$\text{Prox}_{\lambda\alpha_k/(\rho+\theta_{K+1})}^*(\mathbf{Z}) = \mathbf{U} \max(\Sigma - \lambda\alpha_k/(\rho+\theta_{K+1}), 0) \mathbf{V}^\top, \quad (19)$$

where $[\mathbf{U}, \Sigma, \mathbf{V}^\top] = \text{SVD}(\mathbf{Z})$.

$\mathcal{T}$ sub-problem: The $\mathcal{T}$ sub-problem can be formulated as

$$\mathcal{T}^{t+1} = \arg \min_{\mathcal{T}} I_T(\mathcal{T}) + \frac{(K+1)\rho + \theta_{K+2}}{2} \|\mathcal{T} - \mathcal{J}\|_{\text{F}}^2, \quad (20)$$

where its optimal solution is given by

$$\mathcal{T}^{t+1}(i_1, \dots, i_K) = \begin{cases} \mathcal{J}(i_1, \dots, i_K) & \text{if } \mathcal{J}(i_1, \dots, i_K) \in I_T, \\ \varphi & \text{otherwise,} \end{cases} \quad (21)$$

where $\mathcal{J} = (\rho(\mathbf{Y} - \sum_{k=1}^K \mathbf{S}^{k,t} - \mathbf{E}^t) + \mathbf{P}^t + \sum_{k=1}^K (\rho\mathcal{M}^{k,t} - \mathbf{Q}^{k,t}) + \theta\mathcal{T}^t)/((K+1)\rho + \theta_{K+2})$.

$\mathcal{E}$ sub-problem: The $\mathcal{E}$ sub-problem can be formulated as

$$\begin{aligned} \min_{\mathcal{E}} & \frac{\rho}{2} \|\mathbf{Y} - \mathcal{T}^t - \sum_{k=1}^K \mathbf{S}^{k,t} - \mathcal{E} + \frac{1}{\rho} \mathbf{P}^t\|_{\text{F}}^2 \\ & + \frac{1}{2} \|\mathcal{E}\|_{\text{F}}^2 + \frac{\theta_{K+3}}{2} \|\mathcal{E} - \mathcal{E}^t\|_{\text{F}}^2, \end{aligned} \quad (22)$$

where its closed-form solution is given by

$$\mathcal{E} = \frac{\rho(\mathbf{Y} - \mathcal{T}^t - \sum_{k=1}^K \mathbf{S}^{k,t}) + \mathbf{P}^t + \theta\mathcal{E}^t}{1 + \rho + \theta}. \quad (23)$$

$\mathbf{Q}^k, \mathcal{P}$ sub-problems: The variables $\mathbf{Q}^k, k \in [K], \mathcal{P}$ sub-problem can be solved by using the dual ascent method:

$$\begin{cases} \mathbf{Q}^{k,t+1} = \mathbf{Q}^{k,t} + \rho(\mathcal{T}^{t+1} - \mathcal{M}^{k,t+1}), k \in [K], \\ \mathcal{P}^{t+1} = \mathcal{P}^t + \rho(\mathbf{Y} - \mathcal{T}^{t+1} - \sum_{k=1}^K \mathbf{S}^{k,t+1} - \mathcal{E}^{t+1}). \end{cases} \quad (24)$$

Algorithm 1: ADMM-based Optimization Algorithm for ORTRD Model.

input $\mathbf{Y}, \mu, \lambda$.
initialize $\rho = 10^{-5}, \rho_{\max} = 10^{10}, \mu = 1.1, \text{tol} = 10^{-6}$,
 $\mathbf{S}^k = 0, \mathcal{M}^k = \mathcal{T} = \mathbf{Y}, \mathcal{E} = 0$
 1: **while** not converge **do**
 2: # Parallel update for Lines 3 to 4 when a distributed environment is available
 3: Update $\mathbf{S}^{k,t+1}, k \in [K]$, by using Eq. (15).
 4: Update $\mathcal{M}^{k,t+1}$, by using Eq. (18).
 5: # Synchronize $\mathbf{S}^{k,t+1}$ and $\mathcal{M}^{k,t+1}$ when a distributed environment is available
 6: Update $\mathcal{T}^{t+1}$ by using Eq. (21).
 7: Update $\mathcal{E}^{t+1}$ by using Eq. (23).
 8: Update dual variables $\mathbf{Q}^{k,t+1}, k \in [K]$ by Eq. (24).
 9: $\rho = \min(\mu\rho, \rho_{\max})$.
 10: Check the convergence conditions:
 $\min(\text{RelCha}(\mathcal{T}^{t+1}), \text{RelCha}(\mathcal{M}^{k,t+1}), \text{RelCha}(\mathbf{S}^{k,t+1})) \leq \text{tol}$, where $\text{RelCha}(\cdot)$ is defined in Eq. (25).
 11: $t \leftarrow t + 1$.
 12: **end while**
 13: **return** $\mathcal{X}^{t+1}, \{\mathbf{S}^{k,t+1}\}_{k=1}^K$.

The relative change (RelCha) error is utilized to check the algorithm convergence condition, where RelCha is defined by

$$\text{RelCha}(\mathcal{X}) := \frac{\|\mathcal{X}^{t+1} - \mathcal{X}^t\|_{\text{F}}^2}{\|\mathcal{X}^t\|_{\text{F}}^2}. \quad (25)$$

The overall optimization procedures are summarized in Algorithm 1.

C. Convergence Analysis

In this section, we show that the proposed algorithm is guaranteed to converge under a proper selection of regularization parameters $\{\theta_k\}_{k=1}^{K+3}$.

Theorem 1: The sequence $\{\mathbf{S}^{k,t}, \mathcal{M}^{k,t}, \mathcal{T}^t, \mathcal{E}^t\}$ generated by the proposed algorithm converges to a solution $\{\hat{\mathbf{S}}^k, \hat{\mathcal{M}}^k, \hat{\mathcal{T}}, \hat{\mathcal{E}}\}$ to the problem (10). By letting $\gamma > 0$, $\theta_k > \rho(\frac{K+3}{2-\gamma} - 1)\iota_k$, where ι is given by

$$\iota_k = \begin{cases} \sqrt{K}, & k \in [K] \text{ or } k = K+2, \\ 1, & k \in \{K+1, K+3\}, \end{cases} \quad (26)$$

then the sequence $\{\mathbf{S}^{k,t}, \mathcal{M}^{k,t}, \mathcal{T}^t, \mathcal{E}^t\}$ converges to $\{\hat{\mathbf{S}}^k, \hat{\mathcal{M}}^k, \hat{\mathcal{T}}, \hat{\mathcal{E}}\}$.

This theorem establishes that our algorithm achieves convergence when the proximal parameters $\{\theta_k\}$ are properly chosen. The proof details can be found in Section VIII of the supplemental materials.

D. Computational Complexity Analysis

We consider the input tensor size as $d_k = d$, where $k \in [K]$. The computational cost associated with updating $\mathbf{S}^k, \mathcal{E}$, and $\mathcal{T}$

is comprised of simple tensor operations. The principal computational complexity of the proposed Algorithm 1 primarily centers around (18). During each iteration, the update procedure in (18) necessitates performing the singular value decomposition (SVD) on K matrices, each of dimensions $d^{K/2} \times d^{K/2}$, leading to a computational burden of $\mathcal{O}(Kd^{3K/2})$. Consequently, the computation complexity per iteration for Algorithm 1 stands at $\mathcal{O}(d^{3K/2})$.

In the following, we discussed parallelization and approximation strategies to alleviate the computational bottleneck when dealing with large-scale high-order tensors. Note that the proposed ADMM is of the Jacobian type and therefore can be naturally compatible with a distributed implementation without changing the update rules. In a parallel environment, the SVT in the $\mathcal{M}^k$ updates and the $\ell_{2,1}$ proximal operators for $\mathcal{S}^k$ can be computed concurrently across workers or nodes. Therefore, in Algorithm 1, we can distributively update lines 3 to 4 and synchronize them to update lines 6 to 8. In addition, some approximation strategies can also be adopted. In our Algorithm 1, the main computational bottleneck is the SVT operation in (18). The nuclearnorm proximal operator only needs singular values above the softthreshold $\tau_k = \lambda\alpha_k/(\rho + \theta_{K+1})$. Thus, we can replace the full SVD by a rankadaptive randomized SVD (range finder with oversampling p and optional q power iterations) [68] or a Lanczos/PROPACK truncated SVD [69] that computes only the top $\tilde{r}_k$ singular triplets with $\sigma_i > \tau_k$. This approximation can reduce the dominant cost from $\mathcal{O}(d^{3K/2})$ to $\mathcal{O}(Kd^K(\tilde{r}_k+p)(q+1) + Kd^{K/2}(\tilde{r}_k+p)^2)$, where $\tilde{r}_k$ is the numerical rank after thresholding. When the lowTRrank structure is pronounced in data, i.e., $\tilde{r}_k \ll d^{K/2}$, this approximation will lead to substantial computational cost savings.

IV. THEORETICAL RECOVERY PERFORMANCE

To analyze the recovery performance of our approach, we first establish necessary notation. Let $\{\hat{\mathcal{T}}, \hat{\mathcal{S}}, \hat{\mathcal{S}}^k\}$ denote the estimated tensors and $\{\mathcal{T}^*, \mathcal{S}^*, \mathcal{S}^{k,*}\}$ represent their intrinsic true values. We define the corresponding residual tensors as $\Delta\mathcal{T} := \mathcal{T}^* - \hat{\mathcal{T}}$, $\Delta\mathcal{S} := \mathcal{S}^* - \hat{\mathcal{S}}$, and $\Delta\mathcal{S}^k := \mathcal{S}^{k,*} - \hat{\mathcal{S}}^k$. For outlier characterization, we denote $\Omega^k \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$ as the support set of outliers in the k th component $\mathcal{S}^k$, with $\Omega = \sum_{k=1}^K \Omega^k$ representing the overall outlier support set. The term $|\Omega_{(k)}^k|$ indicates the number of corrupted columns in mode k .

Our first theoretical result establishes deterministic bounds on the recovery error:

Theorem 2: If we set the penalty parameters by $\lambda \geq \|\mathcal{E}\|_{trnn}^*$ and $\tau \geq \max_k \|\mathbf{E}_{(k)}\|_{2,\infty}/\beta_k + \max_k 2D\varphi/(\sqrt{d_k}\beta_k) + \max_k (K-1)\varrho/\beta_k$, then any optimal solution of the proposed ORTRD satisfies

$$\begin{aligned} \|\Delta\mathcal{T}\|_F^2 + \sum_{k=1}^K \|\Delta\mathcal{S}^k\|_F^2 &\leq c_1\lambda^2 \sum_{k=1}^K \alpha_k^2 r_k r_{k+s} + c_2\tau^2 |\hat{\Omega}|, \\ \|\Delta\mathcal{T}\|_F^2 + \|\Delta\mathcal{S}\|_F^2 &\leq c_3\lambda^2 \sum_{k=1}^K \alpha_k^2 r_k r_{k+s} + c_4\tau^2 \beta_{k'}^2 |\Omega_{(k')}|, \end{aligned}$$

where $|\hat{\Omega}| = \sum_{k=1}^K \beta_k^2 |\Omega_{(k)}^k|$ and $k' = \arg \min_k \|\mathbf{S}_{(k)}\|_{2,1}$.

Section VII of the supplementary materials contains the full proof. Theorem 2 shows that, once the penalty parameters exceed certain critical values, the recovery errors for both the lowrank tensor $\mathcal{T}$ and the structured sparse tensor $\mathcal{S}$ can be bounded in terms of the latent sparsity levels $|\Omega^k|$ and the TR rank. To establish more precise upper bounds, we must control the $\ell_{2,\infty}$ norm of $\mathbf{E}_{(k)}$ and the spectral norm of the noise tensor $\mathcal{E}$. Therefore, we assume that $\mathcal{E} \in \mathbb{R}^{d_1 \times \dots \times d_K}$ has i.i.d. Gaussian entries $\mathcal{N}(0, \sigma^2)$, which allows us to derive a nonasymptotic upper bound on the estimation error.

Theorem 3: For Gaussian noise tensor $\mathcal{E}$, if the penalty parameters $\lambda = 2\sigma K^{-2} \sum_{k=1}^K (\sqrt{d_{1,k}} + \sqrt{d_{2,k}})/\alpha_k$ and $\tau = \max_k \sigma(\sqrt{d_k} + 3\sqrt{\log D})/\beta_k + \max_k 2D\varphi/(\sqrt{d_k}\beta_k) + \max_k (K-1)\varrho/\beta_k$, then with high probability, any optimal solution of the proposed ORTRD satisfies:

$$\begin{aligned} \|\Delta\mathcal{T}\|_F^2 + \sum_{k=1}^K \|\Delta\mathcal{S}^k\|_F^2 &\leq \frac{c_5\sigma^2}{K^4} \sum_{k=1}^K \left(\frac{\tilde{d}_{1,2}^k}{\alpha_k} \right)^2 \sum_{k=1}^K \alpha_k^2 r_k r_{k+s} + c_6\tau^2 \sum_{k=1}^K \beta_k^2 |\Omega_{(k)}^k|, \\ \|\Delta\mathcal{T}\|_F^2 + \|\Delta\mathcal{S}\|_F^2 &\leq \frac{c_7\sigma^2}{K^4} \sum_{k=1}^K \left(\frac{\tilde{d}_{1,2}^k}{\alpha_k} \right)^2 \sum_{k=1}^K \alpha_k^2 r_k r_{k+s} + c_8\tau^2 \beta_{k'}^2 |\Omega_{(k')}|, \end{aligned} \quad (27)$$

where $\tilde{d}_{1,2}^k = \sqrt{d_{1,k}} + \sqrt{d_{2,k}}, \forall k \in [K]$.

The proof details can be found in Section VII of the supplementary materials. This theoretical result offers several significant advantages over existing approaches. Unlike ORPCA or OTRPCA [28], [70], which rely on strong matrix/tensor incoherence conditions and focus solely on subspace recovery, Theorem 3 provides comprehensive bounds on both the low-rank and outlier components under milder conditions.

For a more intuitive understanding, consider the special case where $d_k = d$, $\alpha_k = 1/K$, and $\beta_k = \beta$. The estimation error bounds in Theorem 3 can be simplified to the following oracle inequality, which holds with high probability:

$$\frac{\|\Delta\mathcal{T}\|_F^2}{D} + \frac{\|\Delta\mathcal{S}\|_F^2}{D} \lesssim \sigma^2 \left(\frac{r^2}{d^{[K/2]}} + \xi^2 \frac{|\Omega_{(k')}|}{D} \right), \quad (28)$$

The notation $a \lesssim b$ signifies that there exists an absolute constant $C > 0$ such that $a \leq Cb$, and we introduce

$$\xi = \sqrt{d} + 3\sqrt{\log D}.$$

This result demonstrates two key theoretical advances over previous work [28], [37], [71]. First, our approach transcends the classical rank limitation of $r \leq d$ in low-rank recovery tasks. Even when $r \geq d$, accurate recovery remains possible provided $r^2 \leq d^{[K/2]}$ —a significant improvement over traditional robust matrix/tensor decomposition methods. Second, while the estimation error of outlier components depends on the minimum number of corrupted columns $|\Omega_{(k')}|$, our method achieves this performance without requiring prior knowledge of the affected modes, matching the performance of mode-specific ORTD approaches [37], [71] while offering greater flexibility.

TABLE II
 COMPARISON WITH CURRENT BASELINES ON THE COMPLEXITY/NON-ASYMPTOTIC BOUND

Methods	Complexity / Non-asymptotic Bound	Verifiable Assumptions	Unknown Outlier Mode	Multiple Outlier Modes	Noise
OR-TPCA [29]	$r_{tubal} \leq C \frac{d_2}{\mu_1 \log d_{(1)}}, \Omega \leq C' d_2$	✗	✗	✗	Slicewise Outliers
RTTD [38]	$\mathcal{O}(r_{tubal} + \Omega^k)$	✓	✗	✗	Elementwise / Fiberwise / Slicewise Outliers + Gaussian Noise
Ours	$\mathcal{O}(r_{TR}^2 + \sum_k \Omega_{(k)}^k)$	✓	✓	✓	Multi-mode Fiberwise Outliers + Subexponential Noise

Remark 2 (Beyond Gaussian): We note that our results are not limited to the Gaussian noise, but can also be extended to a more generalized subexponential noise condition in the following Corollary.

Corollary 1: For subexponential noise tensor $\mathcal{E}$, if the penalty parameters $\lambda = 2\sigma K^{-2} \sum_{k=1}^K (\sqrt{d_{1,k}} + \sqrt{d_{2,k}} + \sqrt{\log(d_{1,k} + d_{2,k}) + \log(d_{1,k} + d_{2,k})})/\alpha_k$ and $\tau = \max_k \sigma (\sqrt{d_k} + \sqrt{\log(D/d_k)} + \log(D/d_k))/\beta_k + \max_k 2D\varphi/(\sqrt{d_k}\beta_k) + \max_k (K-1)\varrho/\beta_k$, then with high probability, any optimal solution of the proposed ORTRD satisfies:

$$\begin{aligned}
 & \|\Delta \mathcal{T}\|_F^2 + \sum_{k=1}^K \|\Delta \mathcal{S}^k\|_F^2 \\
 & \leq \frac{c_5 \sigma^2}{K^4} \sum_{k=1}^K \left(\frac{\tilde{d}_{1,2}^k}{\alpha_k} \right)^2 \sum_{k=1}^K \alpha_k^2 r_k r_{k+s} + c_6 \tau^2 \sum_{k=1}^K \beta_k^2 |\Omega_{(k)}^k|, \\
 & \|\Delta \mathcal{T}\|_F^2 + \|\Delta \mathcal{S}\|_F^2 \leq \\
 & \frac{c_7 \sigma^2}{K^4} \sum_{k=1}^K \left(\frac{\tilde{d}_{1,2}^k}{\alpha_k} \right)^2 \sum_{k=1}^K \alpha_k^2 r_k r_{k+s} + c_8 \tau^2 \beta_{k'}^2 |\Omega_{(k')}^k|, \quad (29)
 \end{aligned}$$

where $\tilde{d}_{1,2}^k = \sqrt{d_{1,k}} + \sqrt{d_{2,k}}, \forall k \in [K]$.

Corollary 1 extends the results in Theorem 3 when the noise follows the subexponential distribution. The results show that by setting λ and τ to an appropriate value (up to a log factor compared to Theorem 3), we achieve a consistent estimation upper bound. Proof of the corollary 1 is given in the Appendix.

We compare our approach with OR-TPCA [28] and RTTD [37] in Table II. Because OR-TPCA provides exact-recovery guarantees, it is not directly comparable in the non-asymptotic column. Our method and RTTD both yield bounds that depend on the tensor's rank and the size of the outlier support. In terms of assumptions, both RTTD and our method require only element-level boundedness (e.g., $\|\mathcal{T}\|_\infty, \|\mathcal{S}_{(l)}^k\|_{2,\infty}$), which is verifiable on real data. Moreover, OR-TPCA and RTTD require the corrupted mode to be known and do not handle simultaneous multi-mode corruption, while our formulation addresses both cases. Finally, regarding noise, OR-TPCA assumes only outliers, RTTD allows outliers plus Gaussian noise, and our analysis accommodates multi-mode outliers plus sub-exponential dense noise. Accordingly, ORTRD adopts more general assumptions on both outliers and dense noise, allowing unknown and multi-mode corruptions and accommodating

subexponential dense noise, thereby surpassing the scope of prior work.

V. EXPERIMENTAL RESULTS

In this section, we present numerical experiments on both synthetic datasets and realworld data. Specifically, light field images and hyperspectral video sequences are adopted to evaluate the efficacy of the proposed method. Our results are compared against several stateoftheart robust matrix and tensor decomposition algorithms: RPCA [72], ORPCA [35], BRTF [73], HoRPCA [14], RTRC [56], TTRPCA [54], ORTPCA [28] and TRGNHAD [32]. The implementation code will be available at <https://github.com/ynqiu/MORTRD> when the final version of this paper is accepted. The experiments are run on MATLAB 2020a on a Linux server with 128 G RAM

The hyperparameters β regarding the sparse component of the tensor-based methods HoRPCA, RTRC, TTRPCA, and the proposed method are set to $\beta_1 = \beta_2 = \dots = \beta_K = 1/(K\sqrt{\max_{k \in [K]}(d_k)})$, and the regularization parameter in ORTPCA are set to $1/\sqrt{d_3 \max(d_1, d_2)}$. The regularization parameter for RPCA and ORPCA is set to $1/\sqrt{\max(d_1, \prod_{k=2}^K d_k)}$ and $1/\sqrt{\log(\prod_{k=2}^K d_k)}$. The low-rank penalty parameter α of TTRPCA is set to [0.2,0.6,0.2], while RTRC and the proposed method are set to [0.35,0.15,0.35,0.15]. The hyperparameter λ is set to 0.1 for the proposed method.

The experimental validation employs multiple metrics, including the clustering accuracy (ACC) for outlier detection, which is defined as

$$\text{clustering accuracy} = \frac{\# \text{ of correctly classified outliers}}{\text{total } \# \text{ of outliers}}.$$

The recovery performance of the low-rank tensor is quantified using the relative squared error (RSE), defined as

$$\text{RSE} = \frac{\|\mathcal{X} - \hat{\mathcal{X}}\|_F^2}{\|\mathcal{X}\|_F^2}. \quad (30)$$

A. Verifying the Correctness of Theorem 3

In this section, we validate the nonasymptotic upper bound established in Theorem 3 through synthetic lowrank tensor recovery experiments. To construct a tensor $\mathcal{T}^* \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$ of TR rank $[r_1, r_2, \dots, r_K]$, we first generate K core tensors $\mathcal{G}^{(k)} \in \mathbb{R}^{r_k \times d_k \times r_{k+1}}$, with each entry sampled independently from $\mathcal{N}(0, 1)$. To construct the latent structural tensor $\mathcal{S}^{k,*}$, we let the support set of $\mathcal{S}_{(k)}^k$ be Ω^k , and then randomly select $|\Omega_{(k)}^k|$

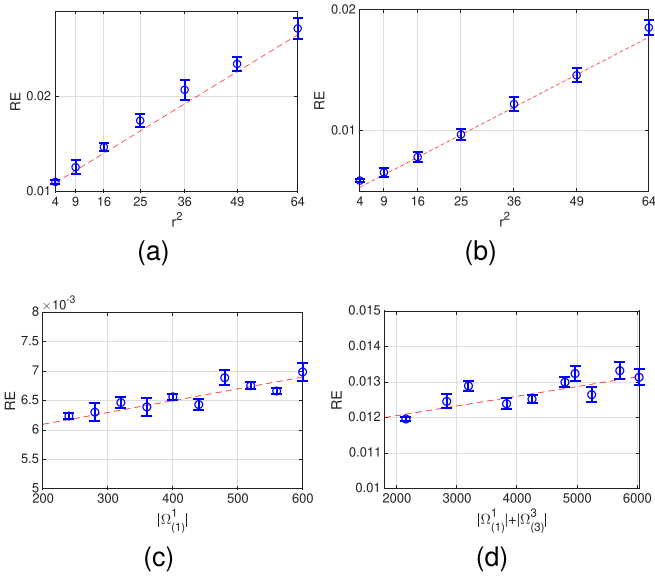


Fig. 3. The relative error (RE) is shown as a function of the number of outliers under two scenarios: (a) corruptions confined to the first mode; (b) corruptions distributed across the first and third modes. Panels (c) and (d) depict the RE versus the squared TR rank for the same mode1 and mode(1,3) corruption settings, respectively.

columns of $\mathbf{S}_{(k)}^k$ as outliers whose entries obey *i.i.d.* $\mathcal{N}(0, 1)$. Thus, the outlier is given by $\mathbf{S} = \sum_{k=1}^K \mathbf{S}^k$. The noise tensor is drawn elementwise from $\mathcal{N}(0, \sigma^2)$ with $\sigma = 10^{-3} \frac{\|\mathcal{T}\|_F}{\sqrt{D}}$, thus maintaining a uniform signal-to-noise ratio. Experiments are replicated ten times, and the mean and standard deviation outcomes are reported.

By (28), the entrywise estimation error of our method grows linearly with the factors $r^2/d^{\lfloor K/2 \rfloor}$ and $|\Omega_{(k')}|/D$. Hence, fixing one factor and increasing the other yields a proportional increase in error. We measure performance using the reconstruction error $RE = \frac{\|\Delta\mathcal{T}\|_F + \|\Delta\mathcal{S}\|_F}{D}$. To assess its dependence on TR rank, we generate a synthetic tensor of size $20 \times 20 \times 20 \times 20$ with $r \in \{2, \dots, 8\}$. Fig. 3(a)–(b) plot RE versus r^2 for outliers in the first mode and in both the first and third modes, respectively, demonstrating the expected linear trend and confirming (28). Fig. 3(c)–(d) display RE against $|\Omega_{(1)}^1|$ and against $|\Omega_{(1)}^1| + |\Omega_{(3)}^3|$, respectively, again showing linear growth with the minimal number of nonzero fibers, in agreement with (27) and (28).

B. Verifying the Effectiveness of Outlier Detection

The approach for generating low-rank and outlier corruptions is the same as the preceding section. The low-rank tensor is of size $20 \times 20 \times 20 \times 20$ with TR rank $r = [3, 3, 3, 3]$. The outlier corruptions are distributed on the first mode $\mathbf{S} = \mathbf{S}^1$ with $|\Omega_{(1)}^1| = 240$. We compute column-wise Frobenius norm of $\hat{\mathbf{S}}_{(1)}^1$ and $\mathbf{S}_{(1)}^{1,*}$, and visualize the first 50 outliers in Fig. 4(a). We can easily find that all the outliers are successfully detected by the proposed approach. Subsequently, we conduct the experiment on multiple latent components by generating outlier tensors $\mathbf{S} =$

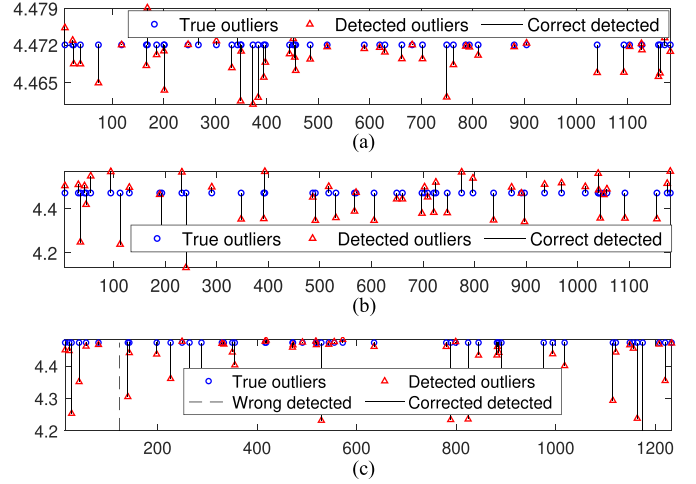


Fig. 4. Column-wise Frobenius norm versus column index. (a) Frobenius norm versus column index when outliers are distributed across the first mode; (b) Frobenius norm versus column index on the first mode when outliers are distributed across the first and the third modes; (c) Frobenius norm versus column index on the third mode when outliers are distributed across the first and the third modes.

$\mathbf{S}^1 + \mathbf{S}^3$ with $|\Omega_{(1)}^1| = |\Omega_{(3)}^3| = 240$. Fig. 4(b)–(c) depict the column-wise Frobenius norm on $\mathbf{S}_{(1)}^1$ and $\mathbf{S}_{(3)}^3$, respectively. All of the outliers in the first mode are successfully detected, but there is one column in $\mathbf{S}_{(3)}^3$ being misclassified as outliers, which demonstrates the effectiveness of the proposed approach in detecting multi-mode outlier corruptions.

C. Hyperspectral Video Recovery

In this experiment, we benchmark the proposed method against state-of-the-art robust matrix / tensor decomposition methods using four hyperspectral video sequences—*bus*, *coin*, *face*, and *hand*—from a hyperspectral object detection dataset.¹ Each sequence is comprised of the first 50 frames captured over 16 spectral bands. To standardize spatial dimensions, every frame is downsampled to $64 \times d_2$, where $d_2 = \text{round}(64 \times \frac{H}{W})$ preserves the original aspect ratio (H, W). Bandwise intensities are scaled to $[0, 1]$, yielding a fourth-order tensor of size $64 \times d_2 \times 16 \times 50$. For matrix-based baselines, the HSV data are matricized along mode1. We corrupt the clean tensor with additive Gaussian noise $\mathcal{N}(0, \sigma^2)$, setting $\sigma = 5 \times 10^{-2} \frac{\|\mathcal{T}\|_F}{\sqrt{D}}$, and introduce sparse outliers by selecting $|\Omega^k| = \text{round}(10^{-2} \prod_{j \neq k} d_j)$ fibers per mode, each assigned a value drawn uniformly from $[-1, 1]$.

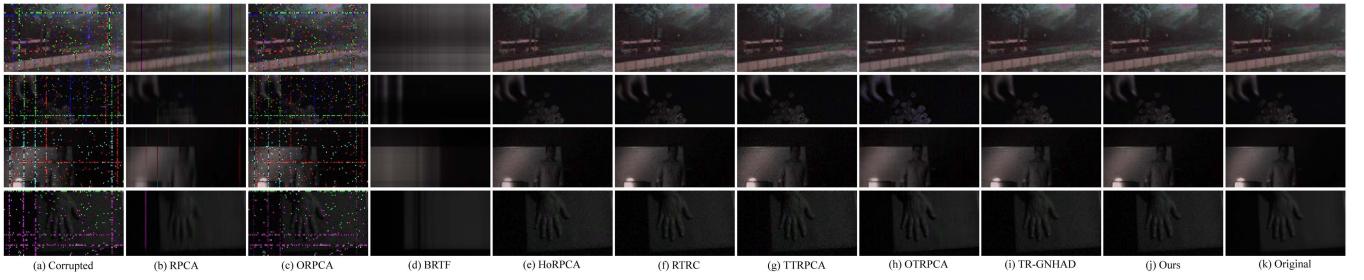
Table III presents the recovery RSE of the compared algorithms on four HSV datasets under various outlier corruption levels. The proposed method consistently achieves the lowest RSE in most scenarios, with the TR-based RTRC method ranking second, thereby validating the advantages of our multimode robust tensor decomposition framework and the fidelity of low-TR-rank representations. Although ORTPCA and ORPCA yield similar results when corruptions are restricted to the first mode, their performance degrades as additional modes are corrupted,

¹<https://www.hsitracking.com/contest/>

TABLE III

 COMPARISON OF RSE METRICS FOR EIGHT METHODS EVALUATED ON FOUR HSV DATASETS, WITH THE LOWEST (BEST) VALUES INDICATED IN **BOLDFACE**

HSV data	Data size	Outlier modes	RPCA	ORPCA	BRTF	HoRPCA	RTRC	TTRPCA	OTRPCA	TRGNHAD	Ours
<i>bus</i>	$64 \times 113 \times 16 \times 50$	1	0.1865	0.0755	0.3152	0.0533	0.0464	0.0523	0.0488	0.0464	0.0452
		1,2	0.1890	0.3712	0.3177	0.0555	0.0473	0.0555	0.0600	0.0468	0.0450
		1,2,3	0.1879	0.5210	0.3191	0.0578	0.0482	0.0587	0.0665	0.0452	0.0451
		1,2,3,4	0.1877	0.6385	0.3199	0.0603	0.0499	0.0637	0.0713	0.0476	0.0452
<i>coin</i>	$64 \times 156 \times 16 \times 50$	1	0.4196	0.1180	0.7105	0.0869	0.0449	0.0522	0.0911	0.0444	0.0417
		1,2	0.4245	0.7354	0.7128	0.0890	0.0458	0.0553	0.1184	0.0423	0.0415
		1,2,3	0.4271	1.0401	0.7140	0.0908	0.0466	0.0585	0.1328	0.0461	0.0416
		1,2,3,4	0.4337	1.2892	0.7153	0.0934	0.0482	0.0637	0.1471	0.0471	0.0416
<i>face</i>	$64 \times 128 \times 16 \times 50$	1	0.1721	0.0773	0.3190	0.0531	0.0458	0.0521	0.0438	0.0427	0.0448
		1,2	0.1743	0.4226	0.3202	0.0548	0.0467	0.0551	0.0557	0.0451	0.0445
		1,2,3	0.1721	0.5953	0.3210	0.0564	0.0475	0.0580	0.0638	0.0465	0.0446
		1,2,3,4	0.1670	0.7308	0.3215	0.0583	0.0487	0.0624	0.0695	0.0477	0.0445
<i>hand</i>	$64 \times 118 \times 16 \times 50$	1	0.1943	0.0710	0.2181	0.0514	0.0461	0.0522	0.0451	0.0420	0.0456
		1,2	0.1915	0.3435	0.2181	0.0529	0.0470	0.0551	0.0516	0.0463	0.0454
		1,2,3	0.1907	0.4823	0.2181	0.0544	0.0479	0.0581	0.0587	0.0469	0.0455
		1,2,3,4	0.1883	0.5908	0.2180	0.0559	0.0489	0.0619	0.0643	0.0478	0.0453


 Fig. 5. The recovered frames of all the compared methods on four HSV datasets (from up to bottom): *bus*, *coin*, *face*, and *hand*. (Best view on a monitor with large resolution).

highlighting the limitation of orientationvariant approaches that depend on precise alignment with column spaces. Conversely, BRTF shows the highest RSE among all methods, which can be attributed to its CPrank criterion that equally penalizes rank across all modes—an assumption that leads to the inherently imbalanced rank characteristics of HSV data (notably often lower in spectral and temporal dimensions, but higher in spatial dimensions).

Fig. 5 depicts the recovered results of all compared methods on four HSV datasets when the HSV datasets are corrupted on (1,2,3,4) modes. The visual comparison results are rather consistent with the quantitative results in Table III. The proposed method can obtain visually superior results than the compared methods. Specifically, the proposed method recovers the HSV data from multi-mode outliers corruptions while finely preserving the HSV structure, while the recovered results of RPCA, ORPCA, HoRPCA, and TTRPCA still retain a small number of outliers.

D. Light Field Image Recovery and Outlier Detection

In this section, we evaluate both the recovery accuracy and the outlier detection performance of all compared algorithms. We employ the publicly available light-field image dataset² and randomly select six images for our experiments. For each light field image, the angular resolution is 9×9 and the spatial

resolution is 512×512 , thus these light field images are naturally expressed as fifth-order tensors of size $512 \times 512 \times 9 \times 9 \times 3$. For computational efficiency, we downsampled the spatial resolution of each light field image to 64×64 and converted it to a grayscale image. Therefore, the tested light field images can be viewed as fourth-order tensors of size $64 \times 64 \times 9 \times 9$. For matrix-based methods, we utilize mode-1 unfolding of the light field images. The outlier corruptions are generated as HSV data recovery section with $|\Omega^k| = \text{round}(10^{-3} \prod_{j,j \neq k} d_j)$. The outlier corruption modes are selected in a candidate set $\{1, [1, 2], [1, 3], [2, 4], [1, 2, 3], [1, 2, 4]\}$. The remaining settings are set as the HSV recovery section.

Fig. 6 presents the RSE values for six lightfield images under different outlier configurations. The proposed method consistently attains the lowest RSE across all cases, demonstrating its robustness; RTRC and TTRPCA yield comparable performance, underscoring the effectiveness of lowTT/TRrank approximations in highorder tensor recovery. As the number of corrupted modes increases, all approaches show modest RSE growth. ORPCA performs competitively when outliers are restricted to the first mode, but degrades under multimode corruption due to its limitation to a single predefined mode.

Subsequently, we investigate the outlier detection results. For the compared methods, we adopt Kmeans clustering to group all $\|\mathbf{S}_{(k)}(:, i)\|_F^2$ into two classes (outliers vs. non-outliers) for outlier detection. The proposed approach automatically separates outliers into different latent components, thus we can directly cluster $\|\mathbf{S}_{(k)}^k(:, i)\|_F^2$ into two classes. We perform

²<https://lightfield-analysis.uni-konstanz.de/>

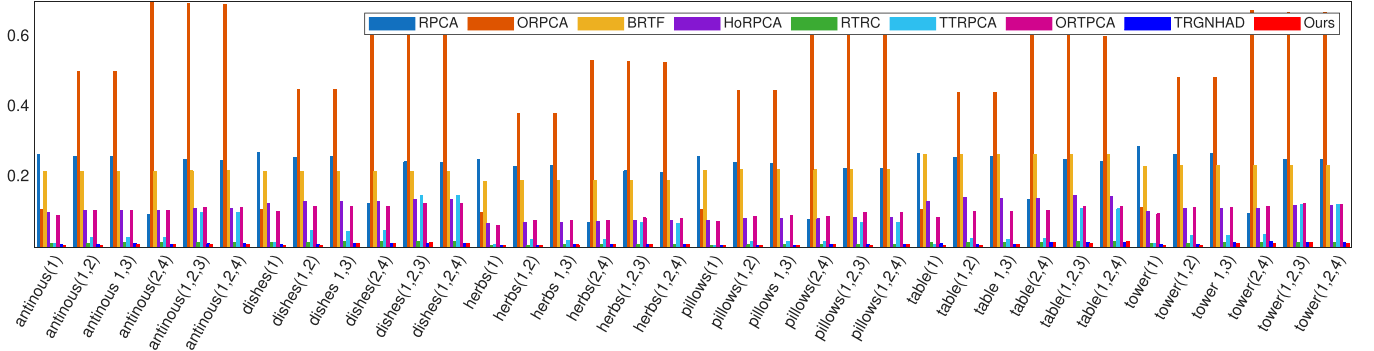


Fig. 6. Comparison of the RSE values of 8 compared methods for image denoising on 6 light field images. The numbers behind the light field images denote the outliers aligned in different modes.

TABLE IV
OUTLIERS DETECTION ACCURACY (%) OF ALL COMPARED METHODS (**BEST IN BOLDFACE**)

Light field data	Outlier modes	RPCA	ORPCA	BRTF	HoRPCA	RTRC	TTRPCA	ORTPCA	TRGNHAD	Ours
<i>antinous</i>	1	100	100	100	100	100	100	100	100	100
	1,2	100	77.48	100	100	100	100	100	100	100
	1,3	89.75	80.64	81.69	81.86	82.04	82.04	82.02	85.58	99.70
	2,4	82.06	94.9	81.83	81.99	82.13	82.13	82.12	86.14	99.66
	1,2,3	83.42	72.53	85.67	85.63	85.55	85.55	85.56	88.66	99.75
	1,2,4	83.49	73.25	85.38	85.34	85.27	85.27	85.28	88.21	99.76
<i>dishes</i>	1	100	100	100	100	100	100	100	100	100
	1,2	100	77.45	100	100	100	100	100	100	100
	1,3	90.92	81.28	81.67	82.37	82.62	82.62	82.59	84.31	99.61
	2,4	82.03	94.77	81.25	82.02	82.13	82.13	82.13	84.81	99.59
	1,2,3	83.46	71.88	86.01	85.68	85.55	85.55	85.58	86.57	99.73
	1,2,4	83.74	73.28	85.66	85.35	85.30	85.30	85.30	86.21	99.70
<i>herbs</i>	1	100	100	100	100	100	100	100	100	100
	1,2	100	77.52	100	100	100	100	100	100	100
	1,3	91.52	81.28	81.40	82.54	82.62	82.62	82.62	85.17	99.65
	2,4	82.07	94.82	81.02	82.01	82.13	82.13	82.12	86.48	99.61
	1,2,3	83.70	72.14	86.09	85.61	85.55	85.55	85.55	88.21	99.72
	1,2,4	83.80	73.83	85.79	85.35	85.30	85.30	85.30	88.68	99.73
<i>pillows</i>	1	100	100	100	100	100	100	100	100	100
	1,2	100	77.57	100	100	100	100	100	100	100
	1,3	90.15	80.87	81.06	81.98	82.17	82.17	82.15	86.71	99.74
	2,4	81.96	94.83	80.99	81.95	82.13	82.13	82.09	86.27	99.63
	1,2,3	83.37	71.67	86.05	85.65	85.55	85.55	85.57	88.69	99.76
	1,2,4	83.36	72.35	85.73	85.35	85.27	85.27	85.29	88.17	99.73
<i>table</i>	1	100	100	100	100	100	100	100	100	100
	1,2	100	77.56	100	100	100	100	100	100	100
	1,3	90.02	81.54	80.60	81.88	82.62	82.62	82.56	85.57	99.57
	2,4	81.45	94.62	80.16	81.43	82.13	82.13	82.04	86.17	99.57
	1,2,3	83.58	72.12	86.39	85.89	85.55	85.55	85.60	88.67	99.69
	1,2,4	83.38	72.47	86.18	85.64	85.30	85.30	85.35	88.18	99.71
<i>tower</i>	1	100	100	100	100	100	100	100	100	100
	1,2	100	77.69	100	100	100	100	100	100	100
	1,3	90.58	80.38	81.42	82.13	82.17	82.17	82.16	84.32	99.64
	2,4	82.10	94.85	81.31	82.08	82.13	82.13	82.12	85.82	99.64
	1,2,3	83.49	71.56	85.85	85.58	85.55	85.55	85.56	86.06	99.70
	1,2,4	83.43	71.85	85.58	85.30	85.27	85.27	85.28	86.61	99.73

Kmeans clustering ten times to mitigate local minima effects, and Table IV reports the mean clustering accuracy across all corrupted modes. Our method consistently achieves the highest accuracy in every scenario, and when outliers are confined to the spatial dimensions (modes 1 and 2), most baseline approaches reach 100% accuracy. To illustrate the mechanisms behind these results, we visualize the estimated sparse outlier tensors for $|\Omega_{(k)}^k| = \text{round}(10^{-3} \prod_{j \neq k} d_j)$, $k \in \{1, 2, 3\}$, reshaping each into a $64 \times 64 \times 81$ thirdorder tensor and displaying them in Fig. 7. The proposed method accurately isolates the true outlier

support, whereas ORPCA fails to recover multimode corruptions and RPCA, BRTF, HoRPCA, and ORTPCA produce sparse components that deviate significantly from the ground truth. These observations confirm the effectiveness of our approach in disentangling outliers arising in multiple modes.

E. Convergence Behaviour

In this part, we examine the convergence behavior of the proposed algorithm on light-field images. Fig. 9(a) shows

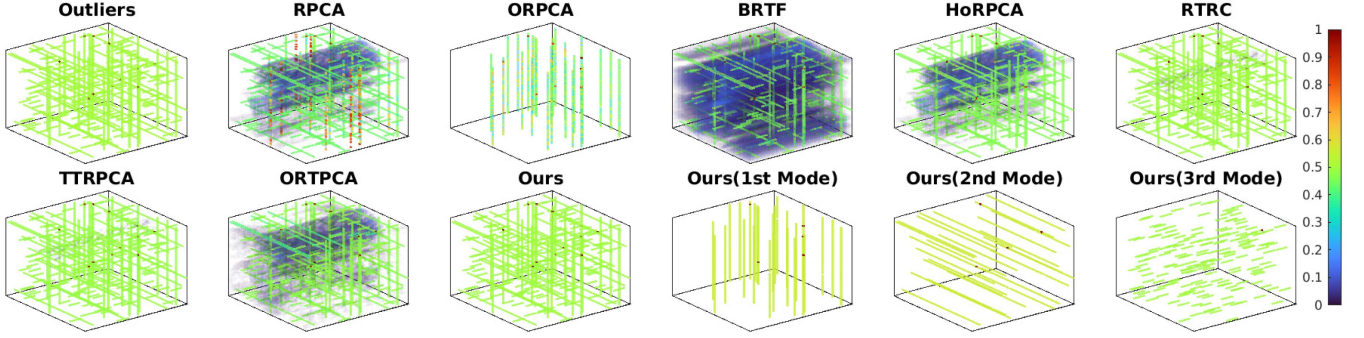
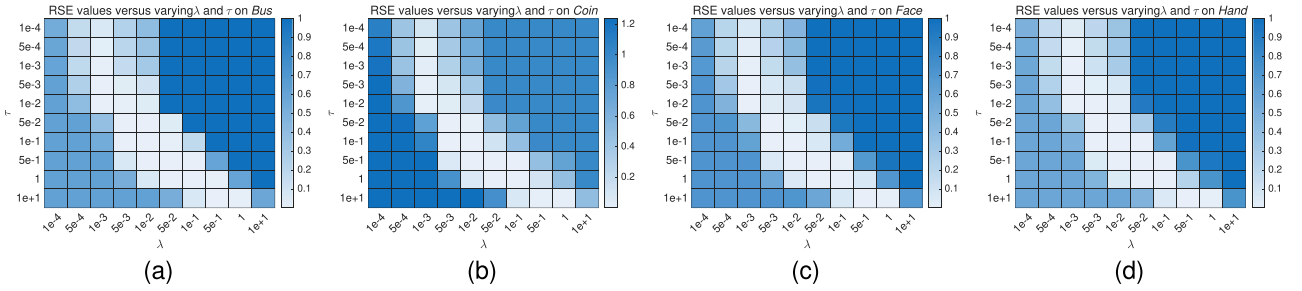
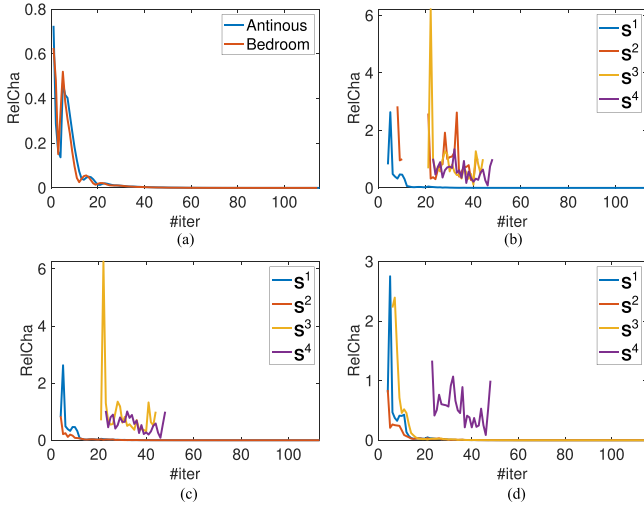


Fig. 7. The visual comparison of the recovered sparse outlier tensors by different methods.


 Fig. 8. Plots of RSE versus varying λ and τ on four HSV data. (a) *Bus*; (b) *Coin*; (c) *Face*; (d) *Hand*.

 Fig. 9. Convergence curves on light-field images. (a) Relative change of the low-rank tensor $\mathcal{T}$ for the *Antinous* and *Bedroom* sequences when outliers occupy modes $\{1, 2, 3\}$; (b)–(d) Relative change of the sparse factors $\{\mathcal{S}^k\}_{k=1}^4$ on the *Antinous* sequence for outlier distributions in $\{1\}$, $\{1, 2\}$, and $\{1, 2, 3\}$, respectively.

the relative change of the recovered low-rank tensor $\mathcal{T}$ for the *Antinous* and *Bedroom* datasets, where $|\Omega_{(k)}^k| = \text{round}(10^{-2} \prod_{j \neq k} d_j)$, $k \in \{1, 2, 3\}$. The results indicate that convergence is typically reached within 50 iterations. Fig. 9(b)–(d) plot the relative change of each sparse component $\mathcal{S}^k$ when outlier modes are $\{1\}$, $\{1, 2\}$, and $\{1, 2, 3\}$, respectively.

In Fig. 9(b), the outliers are only distributed in the first mode. We can observe that the RelCha of the first latent component

$\mathcal{S}^1$ quickly converges closely to zero in around 30 iterations. In contrast, the RelCha of the remaining three latent components $\mathcal{S}^2$, $\mathcal{S}^3$ and $\mathcal{S}^4$ converge to NaN, which indicates that $\mathcal{S}^2 = \mathcal{S}^3 = \mathcal{S}^4 = 0$. Similar phenomena have also appeared in Fig. 9(c) and (d). These findings further validate the algorithm's capability to automatically identify corrupted modes in high-order tensor data.

F. Ablation Study

1) *Hyperparameter Settings*: It is worth noting that, although the proposed method provides a wide range in the context of Theorem 3, we are unable to obtain prior information such as the variance of the noise in practical applications. Therefore, we empirically investigate the sensitivity of parameters λ , τ , and β_k , $k \in [K]$, in HSV recovery task. Since both τ and β_k control the outlier terms, thus we simply let $\beta_k = 1/N * (\max_{k \in [K]}(d_k))$, $k \in [4]$. Due to the space limitation, we show the recovery results on four HSV data by setting the number of outliers $|\Omega^k| = (10^{-2} \prod_{j, j \neq k} d_j)$, $k \in \{1, 2, 3\}$. The hyperparameters λ and τ are selected in a candidate set $\{10^{-4}, 5 \times 10^{-4}, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 10\}$.

Fig. 8 presents the estimation error across a wide range of λ and τ for four HSV data. An over-large λ or τ tends to inflate the estimation error, since they may destroy the balance between low-rank and sparse penalty. In addition, when $\lambda = [2, 100] \times \tau$, the RSE values on four HSV data are consistently low. This linear correlation between λ and τ reduces the burden to adjust hyperparameters. Therefore, we recommend selecting parameters λ and τ within the range of $[0.001, 1]$, while maintaining a linear scaling relationship between them, specifically $\tau = 10\lambda$.

VI. CONCLUSION, LIMITATION AND FUTURE WORK

This paper introduces Outlier Robust Tensor Ring Decomposition (ORTRD), a novel framework that leverages a multi-mode tensor sparsity norm to decompose an observed tensor into a clean lowrank component and a structured sparse outlier component, with provable recovery guarantees under mild assumptions. Comprehensive experiments on both synthetic benchmarks and realworld datasets demonstrate that ORTRD reliably recovers lowrank tensor structure in the presence of multimode corruptions. Some limitations of the current ORTRD should also be noted. The hyperparameters λ and τ should be balanced in an appropriate ratio to avoid suboptimal solutions, as discussed in Section V-F. In addition, although our SVDbased solver for TRNN minimization achieves strong performance on moderatesize problems, its computational complexity may hinder applicability to very large tensors; designing more scalable optimization techniques is a key avenue for future work. Additionally, in many scenarios, data with outliers also contains missing entries. Our current framework does not directly address this case, so developing effective and efficient models for multi-mode outlier-robust tensor decomposition under missing data is a valuable direction to explore.

REFERENCES

- [1] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev.*, vol. 51, no. 3, pp. 455–500, 2009.
- [2] N. D. Sidiropoulos, L. De Lathauwer, X. Fu, K. Huang, E. E. Papalexakis, and C. Faloutsos, "Tensor decomposition for signal processing and machine learning," *IEEE Trans. Signal Process.*, vol. 65, no. 13, pp. 3551–3582, Jul. 2017.
- [3] X. Luo, H. Wu, and Z. Li, "NeuLFT: A novel approach to nonlinear canonical polyadic decomposition on high-dimensional incomplete tensors," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 6148–6166, Jun. 2023.
- [4] F. Di Mauro, K. S. Candan, and M. L. Sapino, "Tensor-train decomposition in the presence of interval-valued data," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 4267–4280, Apr. 2023.
- [5] Z. Wang, H. Nie, W. Zheng, Y. Wang, and X. Li, "A novel tensor learning model for joint relational triplet extraction," *IEEE Trans. Cybern.*, vol. 54, no. 4, pp. 2483–2494, Apr. 2024.
- [6] Y. Panagakis et al., "Tensor methods in computer vision and deep learning," *Proc. IEEE*, vol. 109, no. 5, pp. 863–890, May 2021.
- [7] Y. Qiu, A. Wang, C. Li, H. Huang, G. Zhou, and Q. Zhao, "STEPS: Sequential probability tensor estimation for text-to-image hard prompt search," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 28640–28650.
- [8] Z. Wang, D. Hu, Z. Liu, C. Gao, and Z. Wang, "Iteratively capped reweighting norm minimization with global convergence guarantee for low-rank matrix learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1923–1940, Mar. 2025.
- [9] A. Cichocki et al., "Tensor networks for dimensionality reduction and large-scale optimization: Part 1 low-rank tensor decompositions," *Found. Trends Mach. Learn.*, vol. 9, no. 4–5, pp. 249–429, 2016.
- [10] A. Cichocki et al., "Tensor networks for dimensionality reduction and large-scale optimization: Part 2 applications and future perspectives," *Found. Trends Mach. Learn.*, vol. 9, no. 6, pp. 431–673, 2017.
- [11] J. Liu, C. Zhu, and Y. Liu, "Smooth compact tensor ring regression," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 9, pp. 4439–4452, Sep. 2022.
- [12] Y. Qiu, G. Zhou, Y. Wang, Y. Zhang, and S. Xie, "A generalized graph regularized non-negative Tucker decomposition framework for tensor data representation," *IEEE Trans. Cybern.*, vol. 52, no. 1, pp. 594–607, Jan. 2022.
- [13] H. Huang and C. Ding, "Robust tensor factorization using R 1 norm," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2008, pp. 1–8.
- [14] D. Goldfarb and Z. (Tony) Qin, "Robust low-rank tensor recovery: Models and algorithms," *SIAM J. Matrix Anal. Appl.*, vol. 35, no. 1, pp. 225–253, Jan. 2014.
- [15] Q. Gu, H. Gui, and J. Han, "Robust tensor decomposition with gross corruption," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, vol. 27, pp. 1422–1430.
- [16] X. Pu, H. Che, B. Pan, M.-F. Leung, and S. Wen, "Robust weighted low-rank tensor approximation for multiview clustering with mixed noise," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 3, pp. 3268–3285, Jun. 2024.
- [17] X. Yang, H. Che, and M.-F. Leung, "Tensor-based unsupervised feature selection for error-robust handling of unbalanced incomplete multi-view data," *Inf. Fusion*, vol. 114, no. C., Feb. 2025, Art. no. 102693.
- [18] M. Zhang and C. Ding, "Robust Tucker tensor decomposition for effective image representation," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2013, pp. 2448–2455.
- [19] G. Xu et al., "A robust nonlocal tensor decomposition method for InSAR phase denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5216511.
- [20] W. Cao et al., "Total variation regularized tensor RPCA for background subtraction from compressive measurements," *IEEE Trans. Image Process.*, vol. 25, no. 9, pp. 4075–4090, Sep. 2016.
- [21] Z. Li, Y. Wang, Q. Zhao, S. Zhang, and D. Meng, "A tensor-based online RPCA model for compressive background subtraction," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10668–10682, Dec. 2023.
- [22] X. Zhang, G. Wen, and W. Dai, "A tensor decomposition-based anomaly detection algorithm for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5801–5820, Oct. 2016.
- [23] X. Li et al., "A light-weight and robust tensor convolutional autoencoder for anomaly detection," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 9, pp. 4346–4360, Sep. 2024.
- [24] H. Huang, G. Zhou, Y. Zheng, Y. Qiu, A. Wang, and Q. Zhao, "Adversarially robust deep multi-view clustering: A novel attack and defense framework," in *Proc. 41st Int. Conf. Mach. Learn.*, 2024, pp. 20526–20558.
- [25] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, and S. Yan, "Tensor robust principal component analysis with a new tensor nuclear norm," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 925–938, Apr. 2020.
- [26] Y. Hu and D. B. Work, "Robust tensor recovery with fiber outliers for traffic events," *ACM Trans. Knowl. Discov. from Data*, vol. 15, no. 1, pp. 1–27, 2020.
- [27] Z. Zhang, G. Ely, S. Aeron, N. Hao, and M. Kilmer, "Novel methods for multilinear data completion and de-noising based on tensor-SVD," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 3842–3849.
- [28] P. Zhou and J. Feng, "Outlier-robust tensor PCA," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Honolulu, HI, IEEE, Jul. 2017, pp. 3938–3946.
- [29] Y. Qiu, G. Zhou, A. Wang, Z. Huang, and Q. Zhao, "Towards multi-mode outlier robust tensor ring decomposition," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, pp. 14713–14721.
- [30] Y. Liu, L. Chen, and C. Zhu, "Improved robust tensor principal component analysis via low-rank core matrix," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 6, pp. 1378–1389, Dec. 2018.
- [31] A. Wang, Q. Zhao, Z. Jin, C. Li, and G. Zhou, "Robust tensor decomposition via orientation invariant tubal nuclear norms," *Sci. China Technol. Sci.*, vol. 65, no. 6, pp. 1300–1317, 2022.
- [32] W. Qin, H. Wang, F. Zhang, J. Wang, X. Cao, and X.-L. Zhao, "Tensor ring decomposition-based generalized and efficient nonconvex approach for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539818.
- [33] X. Peng Li, Z.-Y. Wang, Z.-L. Shi, H. C. So, and N. D. Sidiropoulos, "Robust tensor completion via capped Frobenius norm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9700–9712, Jul. 2024.
- [34] M. Yuan and Y. Lin, "Model selection and estimation in regression with grouped variables," *J. Roy. Stat. Soc. Ser. B: Stat. Methodol.*, vol. 68, no. 1, pp. 49–67, 2006.
- [35] H. Xu, C. Caramanis, and S. Sanghavi, "Robust PCA via outlier pursuit," in *Proc. Adv. Neural Inf. Process. Syst.*, 2010, vol. 23, pp. 2496–2504.
- [36] H. Xu, C. Caramanis, and S. Mannor, "Outlier-robust PCA: The high-dimensional case," *IEEE Trans. Inf. Theory*, vol. 59, no. 1, pp. 546–572, Jan. 2013.
- [37] A. Wang, J. Zhong, and G. Tang, "Robust tensor decomposition via t-SVD: Near-optimal statistical guarantee and scalable algorithms," *Signal Process.*, vol. 167, 2020, Art. no. 107319.
- [38] J. D. Carroll and J.-J. Chang, "Analysis of individual differences in multidimensional scaling via an N-way generalization of "Eckart-Young" decomposition," *Psychometrika*, vol. 35, no. 3, pp. 283–319, 1970.
- [39] R. A. Harshman, "Foundations of the PARAFAC procedure: Models and conditions for an "explanatory" multimodal factor analysis," *UCLA Work. Papers Phonetics*, vol. 16, 1970, Art. no. 84.
- [40] L. R. Tucker, "Some mathematical notes on three-mode factor analysis," *Psychometrika*, vol. 31, no. 3, pp. 279–311, 1966.

- [41] I. V. Oseledets, "Tensor-train decomposition," *SIAM J. Sci. Comput.*, vol. 33, no. 5, pp. 2295–2317, 2011.
- [42] Q. Zhao, G. Zhou, S. Xie, L. Zhang, and A. Cichocki, "Tensor ring decomposition," 2016, *arXiv:1606.05535*.
- [43] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, Q. Zhao, and T.-X. Jiang, "Fully-connected tensor network decomposition and its application to higher-order tensor completion," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, pp. 11071–11078.
- [44] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, and D. Meng, "Tensor wheel decomposition and its tensor completion application," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 27008–27020.
- [45] M. E. Kilmer and C. D. Martin, "Factorization strategies for third-order tensors," *Linear Algebra Appl.*, vol. 435, no. 3, pp. 641–658, 2011.
- [46] D. Zhang, H. Huang, Q. Zhao, and G. Zhou, "Generalized latent multi-view clustering with tensorized bipartite graph," *Neural Netw.*, vol. 175, 2024, Art. no. 106282.
- [47] C. Xiao, Y. Huang, H. Huang, Q. Zhao, and G. Zhou, "Consistency and diversity induced tensorized multi-view subspace clustering," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 9, no. 1, pp. 798–809, Feb. 2025.
- [48] A. Wang, Y. Qiu, M. Bai, Z. Jin, G. Zhou, and Q. Zhao, "Generalized tensor decomposition for understanding multi-output regression under combinatorial shifts," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, vol. 37, pp. 47559–47635.
- [49] A. Wang, Y. Qiu, Z. Jin, G. Zhou, and Q. Zhao, "Low-rank tensor transitions (LoRT) for transferable tensor regression," in *Proc. 42nd Int. Conf. Mach. Learn.*, 2025, pp. 64152–64152.
- [50] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, and S. Yan, "Tensor robust principal component analysis: Exact recovery of corrupted low-rank tensors via convex optimization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 5249–5257.
- [51] Y. Qiu, G. Zhou, A. Wang, Q. Zhao, and S. Xie, "Balanced unfolding induced tensor nuclear norms for high-order tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 3, pp. 4724–4737, Mar. 2025.
- [52] B. Huang, C. Mu, D. Goldfarb, and J. Wright, "Provable models for robust low-rank tensor completion," *Pacific J. Optim.*, vol. 11, no. 2, pp. 339–364, 2015.
- [53] Q. Zhao, G. Zhou, L. Zhang, A. Cichocki, and S.-I. Amari, "Bayesian robust tensor factorization for incomplete multiway data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 4, pp. 736–748, Apr. 2016.
- [54] J.-H. Yang, X.-L. Zhao, T.-Y. Ji, T.-H. Ma, and T.-Z. Huang, "Low-rank tensor train for tensor robust principal component analysis," *Appl. Math. Computation*, vol. 367, 2020, Art. no. 124783.
- [55] Y. Qiu, G. Zhou, Z. Huang, Q. Zhao, and S. Xie, "Efficient tensor robust PCA under hybrid model of Tucker and tensor train," *IEEE Signal Process. Lett.*, vol. 29, pp. 627–631, 2022.
- [56] H. Huang, Y. Liu, Z. Long, and C. Zhu, "Robust low-rank tensor ring completion," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1117–1126, 2020.
- [57] A. Cichocki, "Era of Big Data processing: A new approach via tensor networks and tensor decompositions," 2014, *arXiv:1403.2048*.
- [58] J. Yu, C. Li, Q. Zhao, and G. Zhao, "Tensor-ring nuclear norm minimization and application for visual: Data completion," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2019, pp. 3142–3146.
- [59] H. Huang, Y. Liu, J. Liu, and C. Zhu, "Provable tensor ring completion," *Signal Process.*, vol. 171, 2020, Art. no. 107486.
- [60] Y. Qiu, G. Zhou, Q. Zhao, and S. Xie, "Noisy tensor completion via low-rank tensor ring," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1127–1141, Jan. 2024.
- [61] D. Malioutov, M. Cetin, and A. S. Willsky, "A sparse signal reconstruction perspective for source localization with sensor arrays," *IEEE Trans. Signal Process.*, vol. 53, no. 8, pp. 3010–3022, Aug. 2005.
- [62] A. Anandkumar, P. Jain, Y. Shi, and U. N. Niranjan, "Tensor vs. matrix methods: Robust tensor decomposition under block sparse perturbations," in *Proc. Artif. Intell. Statist.*, 2016, pp. 268–276.
- [63] A. Wang, C. Li, J. Zhong, and Q. Zhao, "Robust tensor decomposition via orientation invariant tubal nuclear norms," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 04, pp. 6102–6109.
- [64] L. Jacob, G. Obozinski, and J.-P. Vert, "Group lasso with overlap and graph lasso," in *Proc. 26th Annu. Int. Conf. Mach. Learn.*, 2009, pp. 433–440.
- [65] G. Obozinski, L. Jacob, and J.-P. Vert, "Group lasso with overlaps: The latent group lasso approach," 2011, *arXiv:1110.0413*.
- [66] S. Boyd et al., "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2011.
- [67] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, 2010.
- [68] N. Halko, P.-G. Martinsson, and J. A. Tropp, "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Rev.*, vol. 53, no. 2, pp. 217–288, 2011.
- [69] R. M. Larsen, "Lanczos bidiagonalization with partial reorthogonalization," *DAIMI Rep. Ser.*, vol. 27, no. 537, Dec. 1998, doi: [10.7146/dpb.v27i537.7070](https://doi.org/10.7146/dpb.v27i537.7070). [Online]. Available: <https://tidsskrift.dk/daimipb/article/view/7070>
- [70] H. Xu, C. Caramanis, and S. Sanghavi, "Robust PCA via outlier pursuit," *IEEE Trans. Inf. Theory*, vol. 58, no. 5, pp. 3047–3064, May 2012.
- [71] O. Klopp, K. Lounici, and A. B. Tsybakov, "Robust matrix completion," *Probability Theory Related Fields*, vol. 169, pp. 523–564, 2017.
- [72] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?," *J. ACM*, vol. 58, no. 3, pp. 1–37, 2011.
- [73] Q. Zhao, G. Zhou, L. Zhang, A. Cichocki, and S.-I. Amari, "Bayesian robust tensor factorization for incomplete multiway data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 4, pp. 736–748, Apr. 2016.